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Q19. ∵ 紅點的分佈呈線狀，與直線吻合

∴ The data have been selected from a normal population

Q20. $H_0: \mu = 74$

$H_a: \mu \neq 74$

$$t = \frac{69.7 - 74}{16/\sqrt{25}} = -1.374 > -t_{24,\,0.025} = -2.064$$

⇒ Non-Reject $H_0$. 在 $\alpha = 0.05$ 下，我們沒有足夠的證據證明中風患者的平均CBF與健康人士有顯著差異

Q30. (a) $H_0: \mu = 25$

$H_a: \mu < 25$

$$\frac{20.5 - 25}{5/\sqrt{24}} = -4.3076 < -t_{20,\,0.05} = -1.725$$

$\Rightarrow$ Reject $H_0$, there's sufficient evidence to conclude that the true pretreatment mean is less than 25.

(b) 95% CI $\Rightarrow$ $26.6 \pm t_{20,\,0.025} \times \frac{7.4}{\sqrt{24}} = [23.1315, 29.8685]$

(c) self-esteem score 隨時間增加，代表治療有正面效益，且 s 隨時間增加 $\Rightarrow$ 不同個體在該療程下的效果隨時間增加

Q35. (a) $H_0: \mu = 22$, $H_a: \mu > 22$

$$t = \frac{31-22}{6.2/\sqrt{17}} = 5.9852 > t_{16,\,0.05} = 1.746$$

$\Rightarrow$ Reject $H_0$. there's sufficient evidence to conclude that injected rats drink more water than noninjected rats.

(b) 90% CI. $\Rightarrow$ $31 \pm t_{16,\,0.025} \times \frac{6.2}{\sqrt{17}} = [28.3745, 33.6255]$

$H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 \neq 0$

Q7. $\bar{x}_1 = 7$, $s_1^2 = 15.333$, $\bar{x}_2 = 8.6$, $s_2^2 = 10.3$

$$sp^2 = \frac{(4-1)\times 15.333 + (5-1)\times 10.3}{4+5-2} \approx 12.457 \;\#$$

$$df = 4+5-2 = 7 \;\#$$

$$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{sp^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{7 - 8.6}{\sqrt{12.457\left(\frac{1}{4}+\frac{1}{5}\right)}} \approx -0.7578 > -t_{7,\,0.025} = -2.365$$

第 Q7) R.R. : $|t| > t_{7,0.025} = 2.365$ #

p-value $= 2 \times P[t < -0.6758] \approx 0.52$ # $> 0.05$

$\Rightarrow$ Non-Reject $H_0$，兩母體平均值沒有顯著差異 #

Q8. $S_p^2 = \dfrac{(16-1)\times 4.8 + (13-1)\times 5.9}{16+13-2} \approx 5.2889$ #

$df = 16-1+13-1 = 27$ #

$t = \dfrac{34.6-32.2-0}{\sqrt{5.2889 \times (\frac{1}{16}+\frac{1}{13})}} \approx 2.7949$ # $> t_{27,0.025} = 2.052$

R.R. : $|t| > t_{27,0.025} = 2.052$ #

p-value $= 2 \times P(t > 2.7949) \approx 0.0094$ # $< 0.05$

$\Rightarrow$ Reject $H_0$，兩母體平均值有顯著的差異 #

Q12 (a) $H_0: \mu_1 - \mu_2 = 0$ $H_a: \mu_1 - \mu_2 \neq 0$

$S = \sqrt{\dfrac{8.95^2}{10} + \dfrac{10.72^2}{10}} \approx 4.4113$

$\dfrac{86.4 - 79.7 - 0}{4.4113} \approx 1.5188 < t_{18,0.025} = 2.101$

$\Rightarrow$ Non-Reject $H_0$，$\mu_1, \mu_2$ 沒有顯著的差異

(b) 95% C.I.

$86.4 - 79.7 \pm 2.101 \times 4.4113 = [-2.5681, 15.9681]$

(c) 95% CI 中包含 0 $\Rightarrow$ Non-Reject $H_0$. 我們無法證明使用 ipad 教學比傳統教科書更好 or 更差